

[Collaborative Docs](#) >

Discussion Agenda 08/13/2017

Below is a discussion agenda / live progress report for the Collaboration-wide brainstorming session of the 6th AGORA Workshop @ University of California Santa Cruz (Aug. 11–13, 2017; see the [program](#), [picture](#)).

During this year's workshop, we focus on launching two science-oriented comparisons (see the Paper Workspace links below).

Workshop participants: Everybody should feel free to edit the [meeting notes](#), the content of which will eventually be merged into respective Paper Group Workspace.

Contents

- [1 1. Progress Report \(as of Aug. 2017\)](#)
 - [1.1 \(1\) New Science-oriented Projects](#)
 - [1.2 \(2\) Existing Projects](#)
- [2 2. New Paper Group Ideas](#)
- [3 3. Other Topics for Collaboration-wide Discussion](#)
- [4 4. Recaps](#)
- [5 5. Your Feedback](#)

Check out the important items in red color below!

(1st revision: 08/14/2017, 2nd revision: 08/28/2017)

[1. Progress Report \(as of Aug. 2017\)](#)

We are making progress, chiefly, in the six different fronts below.

[\(1\) New Science-oriented Projects](#)

	Paper "CGM"	Paper "Clumps"	Paper "Metallicity"																																								
GOALS	• Run ~1e12 Ms halo to study absorption lines	• Run ~1e13 Ms halo to study SF clumps at high z	• Run ~1e12 Ms halo to study metal distribution																																								
LEADERS	• Cameron Hummels (Caltech) • Sijing Shen (Oslo)	• Daniel Ceverino (Heidelberg) • Alessandro Lupi (IAP)	•																																								
PARTICIPANTS	• Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Dekel, Mandelker, Primack, Roca-Fabrega, Strawn</td></tr><tr><td>CHANGA</td><td>(Quinn)</td></tr><tr><td>ENZO</td><td>Hummels, Kim, O'Shea, Wise</td></tr><tr><td>GADGET</td><td>Nagamine, Nakamura, Shimizu</td></tr><tr><td>GASOLINE</td><td>Madau, Shen, Wadsley</td></tr><tr><td>GIZMO</td><td>(Dong, Hopkins, Hummels)</td></tr><tr><td>RAMSES</td><td>(Roca-Fabrega, Kretschmer, Teyssier)</td></tr></table> (Code leaders as of Aug. 2017) – ICs help: Buehlmann, Hahn, Simpson – Analysis help: Chakrabarti, Fumagalli,	Code	Contacts	ART-I	Dekel, Mandelker, Primack, Roca-Fabrega, Strawn	CHANGA	(Quinn)	ENZO	Hummels, Kim, O'Shea, Wise	GADGET	Nagamine, Nakamura, Shimizu	GASOLINE	Madau, Shen, Wadsley	GIZMO	(Dong, Hopkins, Hummels)	RAMSES	(Roca-Fabrega, Kretschmer, Teyssier)	• Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Dekel, Ceverino, Mandelker, Primack</td></tr><tr><td>CHANGA</td><td>(Quinn)</td></tr><tr><td>ENZO</td><td>O'Shea</td></tr><tr><td>GADGET</td><td>Nagamine, Nakamura, Shimizu</td></tr><tr><td>GASOLINE</td><td>Mayer, Wadsley</td></tr><tr><td>GIZMO</td><td>Lupi, Hopkins</td></tr><tr><td>RAMSES</td><td>Agertz</td></tr></table> (Code leaders as of Aug. 2017) – ICs help: Buehlmann, Hahn – Analysis help: Chakrabarti, Kim, Turk • Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – XI (high-z galaxies)	Code	Contacts	ART-I	Dekel, Ceverino, Mandelker, Primack	CHANGA	(Quinn)	ENZO	O'Shea	GADGET	Nagamine, Nakamura, Shimizu	GASOLINE	Mayer, Wadsley	GIZMO	Lupi, Hopkins	RAMSES	Agertz	• Code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>GADGET</td><td>Chakrabarti, Lewis</td></tr><tr><td>GEAR</td><td>(Revaz)</td></tr><tr><td>RAMSES</td><td>(Pallottini, Teyssier)</td></tr></table> – Analysis help: Kim • Working Groups engaged: – IV (common analysis) – IX (char. of cos. galaxies) – XII (interstellar medium)	Code	Contacts	GADGET	Chakrabarti, Lewis	GEAR	(Revaz)	RAMSES	(Pallottini, Teyssier)
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	Hummels, Kim, Madau, Roca-Fabrega, Smith, Turk - Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - V (subgrid physics) - IX (char. of cos. galaxies) - X (outflows)		
PROGRESS MADE OR BEING MADE	- Visit Paper's Workspace where Hummels, Shen and others have been updating the progress. - For the homework items (e.g., calibration step #1) to be completed by the telecon in September, please visit the Workspace. - Hummels has provided an ipython notebook that explains how to use yt+Trident on datasets from various codes. - Trident v1.1 available as of Nov. 2017 (see this link; with some preliminary ideas for "demeshened" version of yt and Trident).	- Visit Paper's Workspace where Ceverino, Lupi and others have been updating the progress. - For the homework items (e.g., calibration step #1) to be completed by the telecon in September, please visit the Workspace.	- Visit Paper's Workspace where Chakrabarti and others have been updating the progress. - Will piggyback on the planned Paper "CGM" runs and evolve them further down to $z=0$.
OTHER LINKS	Paper Workspace Trident download and documentation Trident method paper (Hummels et al. 2016)	Paper Workspace	Paper Workspace

(2) Existing Projects

	Paper "DM-Highres"	Paper "NoSubgrid"	Paper "Disk-II"																								
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at higher res	• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Compare simulation codes using common disk ICs																								
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (SDSC)	• Santi Roca-Fabrega (HUJI)																								
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PARTICIPANTS	CHANGA	Quinn, Wallace	ENZO	Hummels, Kim, Simpson, Smith , Turk , Wise	CHANGA	Quinn , Wallace
	ENZO	Hummels, Kim, Simpson , Wise	GADGET	Choi , Nagamine	ENZO	Abel, Butler , Butsky, Forbes, Hummels, Kim , O'Shea, Simpson, Turk, Wise
	GADGET	Choi , Nagamine, Onorbe, (Rocha)	GADGET-SPHS	(Hobbs)	GADGET	Choi , Nagamine, Chakrabarti, Shimizu, (Todoroki)
	GADGET-AFS/CFS	(Hobbs)	GASOLINE	Keller , Madau, Mayer, Shen , Wadsley	GADGET-SPHS	(Hobbs)
	GASOLINE	Madau, Shen, (Pillepich, Zolotov)	GEAR	Revaz	GASOLINE	Benincasa, Keller , Madau, Mayer, Shen, Wadsley
	GCD+	Kawata	GIZMO	Lupi , Thompson	GEAR	Revaz
	GIZMO-AFS	Garrison-Kimmel , Lupi , Hopkins, Onorbe	RAMSES	Agertz, Butler , Teyssier	GIZMO	Lupi , Hopkins, Thompson
	PKDGRAV2	(Guedes, Kuhlen)	(Grackle implementers acknowledged in Paper "4")		RAMSES	Agertz, Butler , DeGraf, Teyssier , Nickerson
	RAMSES	Hahn , Roca-Fabrega, Teyssier			(Code leaders as of Aug. 2016 for Paper "4" before)	
	(Code leaders as of Aug. 2016) – Others interested: Dekel, Feldmann, Kawata, Pallottini, Robles					
PROGRESS MADE OR BEING MADE	• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC. • We may focus on the (survival of) substructures. This subset of topics could potentially be led by Jiang as a smaller paper. • Consistent tree – python interface has been built by Britton Smith (ytree).		• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.		• Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper – previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated. • Analysis items that were not covered in our first disk comparison paper (see this link) will be considered in the next paper, Paper "Disk-II". • A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (pitched in the section below).	
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG III's Workspace. • MUSIC Google group • ytree documentation		• GRACKLE Google group • GRACKLE download & documentation • GRACKLE method paper (Smith et al. 2017) • AGORA API for GRACKLE by Nick Gnedin		• Progress report in Paper's Workspace and/or in WG V's Workspace. • yt-AGORA Google group	

2. New Paper Group Ideas

(1) Metal Diffusion

- Working Group engaged: V (Isolated galaxy),
- Comparison between different codes
- Comparison between different "favorite" metal diffusion methods in different codes
- Using the datasets similar to the ones in our isolated disk comparison (previously known as [Paper "4"](#))
- Can be a subset of what we proposed as Paper "Disk-II" before
- Ideas to be pitched by Roca-Fabrega, Nagamine, Revaz, Kim, etc.

(2) Secular Evolution

- Working Group engaged: V (Isolated galaxy),
- Using the datasets similar to the ones in our isolated disk comparison (previously known as [Paper "4"](#)), but evolved for 1 Gyr with 0.2 Gyr of relaxation step initially
- Can be a subset of what we proposed as Paper "Disk-II" before
- Ideas to be pitched by Roca-Fabrega, Quinn, etc.

(3) Core/Cusp

- Working Group engaged: VII (Dark matter),
- Test models for core/cusp problem, and how this depends on codes and feedback implementations
- **We will piggyback on the planned Paper "CGM" runs -- which satisfies specifications needed for core/cusp studies -- and evolve them further down to $z=0$. We will have to run a high-resolution DM-only run to compare.**
- Ideas to be pitched by Freundlich, Jiang, etc.

3. Other Topics for Collaboration-wide Discussion

(1) Future Funding and Logistics

• At this point, there is no immediate prospect for future funding. Any institution interested in supporting the logistics of future AGORA efforts? U. Zurich is happy to contribute financially to organize a workshop in Zurich, e. g. in 2018 and possibly later. [\[comment by Mayer/Teyssier\]](#) Stanford U. is happy to host small workshops between annual ones, and potentially the annual workshop in 2018 -- but maybe not on weekends. [\[comment by Abel\]](#) Primack et al. will take a look into the new TCAN opportunity for future funding (see [this link](#)). Please let the [project coordinator](#) know if you have any other comments or thoughts.

• As for the long-term needs for the Collaboration, ideally, we would need at least one postdoc with more than one third of his/her time dedicated to AGORA. This is to keep track of various progresses in the Collaboration, to organize meetings and telecons, to analyze and manage the data, etc. [\[thoughts by Kim/Madau/Primack\]](#)

[4. Recaps](#)

(1) AGORA SMBH Prescription for Accretion and Feedback:

- Please visit [this link](#).

(2) Strategy to Store Simulation Outputs:

- Please visit [this link](#).

(3) How to request an account on NERSC:

- If you represent a code group but don't have a NERSC user account, please visit [this page](#). Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

(4) Your Own yt Installation on NERSC:

- Please visit [this link](#).

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit [this link](#).

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)
- Collection of softwares relevant to AGORA effort: [here](#)

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!